

ZESEN LONG

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Incoming computer engineering transfer student and hardware technical lead focused on robotics, embedded controls, and electromechanical prototyping. Experience integrating sensors, actuators, computer vision, CAD/fabrication, and non-blocking firmware in multidisciplinary builds.

EDUCATION

University of California, Irvine — B.S. Computer Engineering; ICS Minor Sep 2026 – Jun 2028
Irvine Valley College — A.A., Computer Science; GPA: 4.00/4.00, multiple Dean's List honors 2023 – 2026

TECHNICAL PROJECTS

- Autonomous Face-Tracking Rover** | *Python, Raspberry Pi, OpenCV, LiDAR, Servo/Motor Control* Spring 2025
- Owned Python control architecture and repository integration for a four-person team, unifying OpenCV face detection, TF-Luna LiDAR ranging, pan/tilt servos, and 4WD motor commands.
 - Engineered closed-loop tracking at an approximately 30 cm following distance; reduced servo jitter with a 5-degree dead band and added robust 9-byte LiDAR frame parsing.
- AI Trash Detector** | *ESP32, XIAO ESP32S3 Sense, TensorFlow Lite, UART, OLED* 2025
- Led development of an edge-AI prototype that classifies waste on-device with a XIAO ESP32S3 Sense and transmits inference results to an ESP32 host over UART.
 - Integrated live class display, counters, buzzer alerts, and serial diagnostics; delivered firmware, model documentation, wiring guidance, demos, and automated repository checks.
- Fan-Adhesion Wall-Climbing Vehicle** | *CAD, 3D Printing, Laser Cutting, Motors, RC* Fall 2025
- Designed and validated a 630 g remotely controlled vehicle using fan-generated pressure differential to drive horizontally and climb a vertical wall.
 - Combined CAD, friction/thrust analysis, laser-cut acrylic, a 3D-printed fan shroud, brushless propulsion, and iterative testing to complete the prototype within a \$100 new-parts budget.
- LEGO-Based Game Console** | *ESP32, Arduino, Embedded C++, OLED, Joystick, Buzzer* 2025 – 2026
- Built Arduino Uno and ESP32 console generations with joystick input, OLED graphics, audio feedback, persistent scores, and fully rewritten multi-game firmware.
 - Designed a CAD-free, reconfigurable enclosure from reusable bricks; evolved the prototype from a Club Day demo into the final reference build for an ESP32 hardware workshop.

LEADERSHIP & ENGINEERING EXPERIENCE

- Irvine Valley College Computer Science Club 2025 – 2026
- First Hardware Technical Lead**
- Designed and led an ESP32 hardware workshop for 20+ students covering board setup, OLED/joystick/buzzer integration, non-blocking firmware, debugging, and hands-on hardware-software interaction.
 - Published reusable materials in a workshop archive and a companion LEGO game-console project to support continued club learning.

SKILLS & OTHER

Programming: Python, C++, Java, MATLAB, Arduino/embedded C++
Robotics & Hardware: Raspberry Pi, ESP32, Arduino, LiDAR, sensors, servos, motor control, breadboarding
Design & Fabrication: SolidWorks, Onshape, 3D printing, laser cutting, friction/thrust analysis, iterative testing
Software & Tools: Git/GitHub, OpenCV, TensorFlow Lite, Verilog, Multisim